

Test worksheet

Generated: May 28, 2026 | Questions: 3 Items

Worksheet Question Index Checklist

- ☐ Question 1: VCE 2025 Specialist Maths Exam 2 · Section B Q3 (10 mk)
- ☐ Question 2: VCE 2025 Specialist Maths Exam 2 · Section A Q6 (1 mk)
- ☐ Question 3: VCE 2025 Specialist Maths Exam 2 · Section B Q1 (10 mk)

TOTAL MARKS: 21 Marks Available

Question 3 (10 marks)

A tank initially contains 5 kg of salt dissolved in 3000 litres of water. Salty water that contains 0.1 kg of salt per litre of water enters the tank at a rate of 20 litres per minute. The solution is kept thoroughly mixed and drains from the tank via a tap at the same rate of 20 litres per minute.

- a.** By considering concentration, explain whether the quantity of salt in the tank increases with time.

1 mark

- b.** Let Q denote the quantity of salt, in kilograms, in the tank at time t minutes.

Show that Q satisfies the differential equation $\frac{dQ}{dt} = \frac{300 - Q}{150}$.

1 mark

- c.** Using Euler's method with a step size of 15 minutes, find $Q(30)$, the approximate quantity of salt in the tank after 30 minutes.

Give your answer in kilograms, correct to two decimal places.

2 marks

- d. Use calculus to solve the differential equation $\frac{dQ}{dt} = \frac{300-Q}{150}$, expressing Q in terms of t . 3 marks

- e. What value does the quantity of salt in the tank approach as time approaches infinity?
Give your answer in kilograms. 1 mark

- f. Find the time taken for the quantity of salt in the tank to reach 100 kg. 1 mark

- g. When the quantity of salt in the tank reaches 100 kg, the tap draining the tank is turned off. Assume that the tank does not overflow and there is no change to the inflow rate.
After the tap is turned off, how many minutes does it take for the concentration of salt in the tank to reach $\frac{1}{20}$ kg L⁻¹? 1 mark

Question 5

The equation $z^3 + az^2 + bz - 52 = 0$, where $a, b \in \mathbb{R}$ and $z \in \mathbb{C}$, has a solution $z = 2 - 3i$.

The value of ab is

- A. -232
- B. -64
- C. -8
- D. 0

Question 6

Let $z \in \mathbb{C}$.

Given that $|z| = 1$ and $z \neq 1$, $\operatorname{Re}\left(\frac{1}{1-z}\right)$ is

- A. $-\frac{1}{2}$
- B. 0
- C. $\frac{1}{2}$
- D. $\frac{\sqrt{3}}{2}$

Question 7

Using the substitution $u = \cos(\theta)$, $\frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin(2\theta)}{1 + \cos(\theta)} d\theta$ can be expressed as

- A. $\int_0^{\frac{\pi}{2}} u \sqrt{\frac{1-u}{1+u}} du$
- B. $\int_0^1 \left(1 + \frac{1}{1+u}\right) du$
- C. $\int_0^{\frac{\pi}{2}} \left(1 - \frac{1}{1+u}\right) du$
- D. $\int_0^1 \left(1 - \frac{1}{1+u}\right) du$

Section B

Instructions

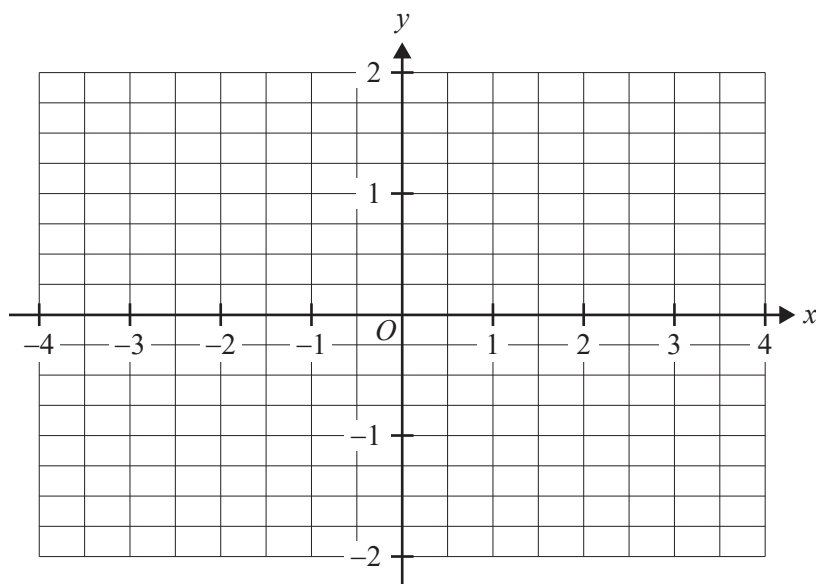
- Answer **all** questions in the spaces provided.
- Write your responses in English.
- Unless otherwise specified, an **exact** answer is required for each question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude $g \text{ m s}^{-2}$, where $g = 9.8$

Question 1 (10 marks)

- a. Sketch the graph of $y(x) = \frac{3x}{x^3 + x + 2}$ on the axes below.

Label the asymptotes with their equations, and label the turning point and the point of inflection with their coordinates. Give the coordinates of the point of inflection correct to one decimal place.

3 marks



- b. The region bounded by the graph of $y = \frac{3x}{x^3 + x + 2}$, the coordinate axes and the line $x = 2$ is rotated about the x -axis to form a solid of revolution.

- i. Write down a definite integral that, when evaluated, will give the volume of the solid of revolution.

1 mark

ii. Find the volume of the solid of revolution correct to two decimal places.

1 mark

c. Find the equations of the vertical asymptotes of the curve given by $y = \frac{3x}{x^3 - 5x + 2}$.

1 mark

d. A family of curves is given by $y(x) = \frac{3x}{x^3 + ax + 2}$, where $a \in \mathbb{R}$.

i. Consider the case where the graph has a stationary point P .

Find the y -coordinate of P in terms of a .

1 mark

ii. For a given value of a , the graph has no stationary points.

Find the equations of the vertical asymptotes of the graph in this case.

1 mark

iii. For a given value of a , the graph will have a point of inflection at $x = 2$.

Find the value of a .

2 marks
